

Jorge Baltazar Acuña Vallejos

Data Analyst / Data Scientist

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Data professional with 3+ years building ETL/ELT pipelines, Power BI dashboards, and ML-driven analytics (anomaly detection, predictive modeling) to integrate multi-source data into reliable, governed, automated reporting systems, backed by 8 years of professional experience that began in mechanical engineering. Skilled at building automated data pipelines, architecting relational databases, and translating complex datasets into executive dashboards. Certified Scrum Master (SMPC) with practical experience in Product Ownership, aligning technical analysis with business and product goals.

SKILLS

- **Programming & Scripting:** Python, SQL, R (tidyverse), JavaScript, VBA, HTML, CSS.
- **Data Engineering & Cloud:** Google BigQuery, PostgreSQL, Firebase, ETL Pipelines, Modern Data Stack.
- **Data Science & Analytics:** Machine Learning (XGBoost, SVM), A/B Testing, Statistical Analysis, Predictive Modeling, Operations Research (PuLP).
- **Visualization & BI:** Power BI, Tableau, Google Looker Studio, Streamlit, Advanced Excel.
- **Agile & Methodology:** Scrum Master (SMPC/SFPC), Product Management, Cross-functional Leadership.

WORK EXPERIENCE

R&D Engineer

Procure, Santiago, Chile, June 2022 to April 2025

- Built an ML-driven anomaly detection model (Python, SVM) that flagged system-performance deviations up to 4 hours in advance, enabling proactive maintenance and reducing unplanned downtime.
- Designed and maintained ETL/ELT pipelines (Python, VBA) integrating multi-source corporate data into centralized reporting systems, automating report generation and cutting processing time by 50% while improving data traceability.
- Built and maintained Power BI dashboards with structured data models and DAX measures, delivering automated, real-time reporting that shaped executive decision-making and reduced manual report preparation.
- Led cross-functional Agile teams as Scrum Master to build and launch a digital project management web application, implementing geographic visualizations that drove product adoption.
- Owned end-to-end data architecture (Firebase, BigQuery) for a production software product, implementing data governance practices — schema standards, validation rules, and quality checks — to ensure consistency and traceability across integrated reporting systems.
- Data Quality & Warehousing: Evolved and maintained schemas (Firestore) by adding columns, adjusting data types, and validating external CSV/text sources through custom scripts before ingestion, ensuring data integrity across reporting pipelines.
- User & Stakeholder Training: Trained end users on software adoption and educated business stakeholders on BI dashboard usage and underlying data collection methodology, improving self-service reporting and data literacy.

Design Engineer

Flutek, Santiago, Chile, January 2021 to June 2022

- Processed and filtered millions of data points from high-precision spatial data acquisition tools, transforming high-density point clouds into standardized digital models.
- Conducted complex data reconciliation between physical inventories and digital 3D models, identifying deviations and ensuring 100% data accuracy for a conceptual Digital Twin system.

Design Engineer

Neptuno Pumps, Iquique, Chile, May 2018 to January 2021

- Processed and analyzed empirical performance data from physical bench tests, comparing results against theoretical baselines to validate system reliability and data accuracy.
- Delivered highly optimized, data-driven technical proposals utilizing rigorous statistical analysis, significantly increasing solution efficiency and ROI for high-value B2B clients.

Project Engineer

Minera Valle Central, Rancagua, Chile, March 2017 to May 2018

- Built Advanced Excel models utilizing Pivot Tables and Macros to transform raw operational data into dynamic, actionable management reports for senior leadership.
- Extracted and normalized time and resource data to rigorously track KPIs, integrating real-time data to update projection models and keep operational projects strictly under budget.

PROJECTS

Rx Commercial Intelligence

Python, Snowflake, SQL, XGBoost, Power BI, SciPy

- Architected a scalable star schema in Snowflake and engineered a robust Python ETL pipeline (pandas) to process multi-source pharmaceutical data resulting in a governed, analytics-ready cloud data warehouse that ensured complete data integrity and eliminated duplicate grain violations.
- Trained a gradient boosting predictive model (XGBoost) utilizing engineered features like market-access state and historical field-force intensity resulting in highly accurate, territory-level next-quarter demand forecasts (validated via MAPE) that fed directly into executive reporting.
- Executed a rigorous field-force ROI analysis applying a difference-in-differences statistical framework and significance testing (t-test / Mann-Whitney) resulting in the isolation of true causal revenue lift and the delivery of a mathematically backed investment recommendation for commercial leadership.
- Developed a comprehensive 5-page Power BI executive dashboard featuring custom DAX measures connected directly to the Snowflake SQL views resulting in an automated, single source of truth that allowed stakeholders to instantly track reimbursement funnels, gross-to-net revenue waterfalls, and territory performance.

Aero-Optimize: Predictive & Prescriptive Airline Operations System

Python, SQL, Scikit-Learn, PuLP, Streamlit, GitHub

- Architected an end-to-end machine learning pipeline extracting and transforming large-scale aviation datasets via PostgreSQL to train gradient boosting models (XGBoost), resulting in a scalable backend that successfully forecasts flight delay probabilities.
- Engineered a prescriptive Operations Research optimization model utilizing Mixed-Integer Programming (PuLP) to dynamically reassign flight crews during disruptions, resulting in minimized operational downtime and cascading costs without violating strict FAA safety constraints.
- Deployed the predictive engine as a production-ready, interactive web application utilizing Streamlit and FastAPI, resulting in a centralized digital product that directly enables business stakeholders to execute instantaneous, data-driven operational decisions.

Cloud ETL & Digital Marketing Performance Dashboard

Python, BigQuery, Google Looker Studio, GitHub

- Engineered a cloud-based Modern Data Stack pipeline extracting and transforming raw marketing data using Python to load into Google BigQuery.
- Conducted rigorous statistical A/B testing (Z-Test, Chi-Square) and calculated precise financial KPIs (ROAS, CPA) to evaluate campaign effectiveness.
- Designed an executive-level Looker Studio dashboard featuring automated budget-alert logic and dynamic filtering to instantly identify and mitigate unprofitable ad spend.

End-to-End Solar Energy Dashboard

Python, PostgreSQL, Tableau, GitHub

- Engineered a data pipeline utilizing Python to programmatically extract and process historical solar radiation data via the NASA POWER API, resulting in a centralized PostgreSQL relational database serving as a single source of truth.
- Developed a highly interactive Tableau dashboard utilizing complex calculated fields and parameter actions, resulting in the dynamic injection of technical hardware specifications and financial data directly from the SQL database.
- Designed a custom, mobile-first UI/UX featuring smart geographic filtering and mathematical aggregations, resulting in a digital tool that enables users to accurately estimate solar panel requirements and turnkey financial investments based on real-time localized weather data.

EDUCATION

- **Bachelor of Science in Mechanical Engineering (M.S. Equivalent), 2009 to 2015**
Pontificia Universidad Católica de Chile
- **Sustainable Energies Diploma, 2017**
Pontificia Universidad Católica de Chile

CERTIFICATIONS

- **Google Data Analytics Professional Certificate, 2025**
- **Scrum Master Professional Certificate (SMPC), 2023**

LANGUAGES

- **Spanish:** Native
- **English:** C1
- **Italian:** B1
- **Portuguese:** B1